

Hye Sun Yun

Computer & Data Science
Lafayette College

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RESEARCH INTERESTS

Human-AI Interaction, Responsible AI, Human-Computer Interaction, Natural Language Processing, & Healthcare

ACADEMIC POSITIONS

Lafayette College	Fall 2026–Present
<i>Assistant Professor</i>	<i>Easton, PA</i>
<i>Department of Computer Science & Program of Data Science</i>	

EDUCATION

Northeastern University	2020–2026
<i>Ph.D. in Computer Science</i>	<i>Boston, MA</i>
<i>Thesis: LLMs for Consumer Health Information:</i> <i>Understanding Users and Models for Responsible Health AI</i> <i>Committee: Byron C. Wallace, Timothy Bickmore, Mai ElSherief, & Emma Pierson</i>	
Northeastern University	2024
<i>M.S. in Computer Science</i>	<i>Boston, MA</i>
Wellesley College	2013–2017
<i>B.A. in Computer Science and Africana Studies</i>	<i>Wellesley, MA</i>
<i>Magna Cum Laude; Phi Beta Kappa</i>	

TEACHING EXPERIENCE

Lafayette College

Introduction to Computer Programming (CS110)	Fall 2026
1 lecture section, 24 students, 1 TA	
Principles of Data Science (DS 201)	Fall 2026
1 lecture section, 15 students	

Northeastern University

Intermediate Programming with Data (DS2500) – with Deahan Yu	Fall 2025
1 out of 4 lecture sections, 240 students for course, 14 TAs	
Special Topics: Research in Human-Centered NLP (CS4973/6983)	Spring 2025
1 lecture section, 10 students	

Teaching Assistant

- Introduction to CS Research (CS3950) at Northeastern University Fall 2024
- Quality Assurance to Quality Engineer Program at Wayfair Fall 2019
- MVC Framework Course at Wayfair 2019–2020
- Tangible User Interfaces (CS320) at Wellesley College Fall 2016
- Languages & Automata (CS235) at Wellesley College Fall 2016
- Algorithms (CS231) at Wellesley College Spring 2016
- Beginning Japanese (JPN101,102) at Wellesley College 2014–2015

Workshops

- Introduction to GitHub (Khoury SHEROs of Color Hackathon)** March 2025
Instructor at Northeastern University

Guest Lectures

- “Can We Trust AI to Answer Our Health Questions?”** Spring 2026
Course: CS3950 Introduction to CS Research at Northeastern University
- “Virtual Agents for Health Behavior Change”** Fall 2022
Course: CS3950 Introduction to CS Research at Northeastern University

PUBLICATIONS

Preprints

1. **Hye Sun Yun**, Geetika Kapoor, Michael Mackert, Ramez Kouzy, Wei Xu, Junyi Jessy Li, & Byron C. Wallace. [“This Treatment Works, Right? Evaluating LLM Sensitivity to Patient Question Framing in Medical QA”](#) In Review, 2025.

Peer-Reviewed Conference Papers

1. **Hye Sun Yun**, Karen Y.C. Zhang, Ramez Kouzy, Iain J. Marshall, Junyi Jessy Li, & Byron C. Wallace. [“Caught in the Web of Words: Do LLMs Fall for Spin in Medical Literature?”](#) In *Proceedings of Conference on Health, Inference, and Learning (CHIL)*, 2025.
2. **Hye Sun Yun** & Timothy Bickmore. [“Framing Health Information: The Impact of Search Methods and Source Types on User Trust and Satisfaction in the Age of LLMs.”](#) In *Proceedings of the Extended Abstracts of the CHI Conference*, 2025.
3. **Hye Sun Yun**, David Pogrebitskiy, Iain J. Marshall, & Byron C. Wallace. [“Automatically Extracting Numerical Results from Randomized Controlled Trials with Large Language Models.”](#) In *Proceedings of the Machine Learning for Healthcare (MLHC) Conference*, 2024.
4. **Hye Sun Yun**, Mehdi Arjmand, Phillip Sherlock, Michael K Paasche-Orlow, James W Griffith, & Timothy Bickmore. [“Keeping Users Engaged During Repeated Interviews by a Virtual Agent: Using Large Language Models to Reliably Diversify Questions.”](#) In *Proceedings of the ACM International Conference on Intelligent Virtual Agents (IVA)*, 2024.

5. Mina Fallah, Farnaz Nouraei, **Hye Sun Yun**, & Timothy Bickmore. “Investigating User Perceptions of Collaborative Agenda Setting in Virtual Health Counseling Session.” In *Proceedings of the ACM International Conference on Intelligent Virtual Agents (IVA)*, 2024.
6. **Hye Sun Yun**, Iain J. Marshall, Thomas A. Trikalinos, & Byron C. Wallace. “Appraising the Potential Uses and Harms of LLMs for Medical Systematic Reviews.” In *Main Proceedings of the Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2023.
7. Prasanth Murali, Ian Steenstra, **Hye Sun Yun**, Ameneh Shamekhi, & Timothy Bickmore. “Improving Multiparty Interactions with a Robot Using Large Language Models.” In *Proceedings of the Extended Abstracts of the CHI Conference*, 2023.
8. **Hye Sun Yun**, Matias Volonte, & Timothy Bickmore. “Motivating health behavior change with a storytelling virtual agent.” In *Proceedings of the ACM International Conference on Intelligent Virtual Agents (IVA)*, 2022.

Journal Articles

1. **Hye Sun Yun** & Timothy Bickmore. “Online Health Information–Seeking in the Era of Large Language Models: Cross-Sectional Web-Based Survey Study.” *Journal of Medical Internet Research (JMIR)*, Volume(27), 2025.
2. Iain J. Marshall, Thomas A. Trikalinos, Frank Soboczenski, **Hye Sun Yun**, Gregory Kell, Rachel Marshall, & Byron C. Wallace. “In a pilot study, automated real-time systematic review updates were feasible, accurate, and work-saving.” *Journal of Clinical Epidemiology*, Volume(153), 2023.

Workshop Papers

1. **Hye Sun Yun**, Shuo Zhou, Everlyne Kimani, Stefan Olafsson, Teresa K. O’Leary, Dhaval Parmar, Jessica A. Hoffman, Stephen S. Intille, Michael K. Paasche-Orlow, & Timothy Bickmore. “Techno-spiritual Engagement: Mechanisms for Improving Uptake of mHealth Apps Designed for Church Members.” In *HEALTHI at Conference on Intelligent User Interfaces (IUI)*, 2022.

Media Articles

1. Teresa K. O’Leary, Elizabeth Stowell, Darley Sackitey, **Hye Sun Yun**, David Wright, Michael K. Paasche-Orlow, Timothy Bickmore, & Andrea G. Parker. “Church after Sunday: supporting everyday well-being through techno-spiritual health interventions.” In *ACM Interactions*, 2022.

PROFESSIONAL EXPERIENCE

Truveta

AI/ML Research Intern

Summer 2024

Remote (Seattle, WA)

Mentors: Saman Zarandioon, Mehraveh Salehi, Sadra Naddaf

- Conducted research in the field of LLMs and clinical notes investigating quality and

diversity of training data for clinical concept extraction

- Explored methods to optimize a generalized clinical concept extraction LLM

Wayfair

2017–2020

Full-Stack Software Engineer/Manager

Boston, MA

- Developed and implemented features for a supplier-facing web tool, enabling efficient bulk addition of new products to the catalog
- Successfully spearheaded two major, multi-phase refactoring initiatives: frontend migration from Tungsten to React and Redux and completely rewriting the team's Excel sheet import service
- Provided guidance and mentorship during rapid team growth, facilitating the onboarding of 10+ new hires
- Led team-wide adoption of pair programming to foster collaboration and knowledge sharing and established a program to train new code reviewers
- Managed and led a team of three software engineers, cultivating a collaborative and productive work environment, and ensuring successful project delivery

RESEARCH MENTORSHIP

Karen Y.C. Zhang , B.S. in Computer Science at <i>Northeastern University</i>	Jan–Oct 2025
Patrick Wilson , Master's in Public Health at <i>King's College London</i>	Spring–Fall 2024
David Pogrebetskiy , B.S. in Data Science at <i>Northeastern University</i>	Spring 2024
Nikita Salkar , M.S. in Computer Science at <i>Northeastern University</i>	Spring 2022

HONORS & AWARDS

Khoury College PhD Teaching Award Khoury College of Computer Sciences, <i>Northeastern University</i>	2026
Selected as a mentee for CSST Summer Institute 2025 Consortium for the Science of Sociotechnical Systems, <i>Syracuse University</i>	2025
Selected as a participant for CRA-WP Grad Cohort for Women Computing Research Association–Widening Participation, <i>San Francisco, CA</i>	2023
Best Rapid-Fire Presentation Award PRISM Health Symposium, <i>San Francisco, CA</i>	2021

SERVICE & OUTREACH

Northeastern University Institutional Service

PhD Open House <i>Student Panelist & Prospective Student Buddy</i>	2021 & 2024
Pathway to PhD Workshop	Summer 2022

PhD Student Panelist

Wellesley College Research Students' Visit

Summer 2022

PhD Student Panelist

PhD Hiring Committee

2022

Faculty Application Reviewer

PhD CS Admissions Committee

2021

PhD Student Application Reviewer

Community Engagement

Code to PhD

Fall 2025-Present

PhD Application Support Program Mentor

Khoury SHEROs of Color

Spring 2025

PhD Student Panelist & Hackathon Mentor

Queer in AI

Fall 2023

Graduate School Application Review Program Mentor

Wayfair's Women+ in Software

2019–2020

Co-founder & Leader

Peer Reviewer

Conferences

CHI 2023, 2025, & 2026

Machine Learning for Healthcare (MLHC) 2025–2026

Conference on Language Modeling (COLM) 2025–2026

IEEE Virtual Reality 2025

Intelligent Virtual Agents (IVA) 2023–2026

Autonomous Agents and Multiagent Systems (AAMAS) 2023

IEEE International Symposium on Mixed and Augmented Reality (ISMAR) 2023

Journals

International Journal of Human-Computer Studies 2025

Journal of Medical Internet Research (JMIR) 2025–2026

Journal of Health Communication 2026

Cochrane Evidence Synthesis and Methods 2025

Digital Health (SAGE) 2026

Computers in Human Behavior Reports 2026

Research Synthesis Methods 2024–2025

Workshops

ACL Workshop on Human-Centered Large Language Modeling 2024

NeurIPS Workshop on Robustness in Sequence Modeling 2022

ACM Web Conference Workshop on UserNLP 2022

TALKS

1. [“Beyond Hallucinations: Unveiling Hidden Dangers of LLMs in Health Information Access”](#) (June 2025). Invited talk. Wellesley College.
2. [“Beyond Hallucinations: Unveiling Hidden Dangers of LLMs in Health Information Access”](#) (April 2025). Invited talk at Department Seminar. Clemson University.
3. [“Charting My CS Career Path: From Wellesley to Industry and PhD”](#) (Nov. 2024). Invited talk. Wellesley College Hackathon.
3. “Automatically Extracting Numerical Results from RCTs with LLMs” (Oct. 2024). Invited talk. [Codamatrix](#) Reading Group.
4. “Using LLMs to Maintain User Engagement in Repeated Virtual Agent Interviews” (Sept. 2024). Conference talk. IVA 2024.
5. “Techno-spiritual Engagement: Mechanisms for Improving Uptake of Church mHealth App” (March 2022). Workshop talk. IUI HEALTHI Workshop 2022.
6. “Longitudinal Analysis of African-American Vaccine Attitudes on Social Media” (Dec. 2021). Rapid Research Talk with Silvio Amir. PRISM Health Symposium. **[Best Rapid-Fire Presentation Award]**

PRESS COVERAGE

1. [“As AI expands into medicine, Northeastern study finds AI models influenced by medical bias”](#) by Emily Spatz. *Khoury News*, August 2025.
2. [“Dr. Google Starts Sharing Regular Folks’ Advice As Chatbots Loom”](#) by Michael L. Millenson. *Forbes*, May 2025

PROFESSIONAL DEVELOPMENT

Graduate Postsecondary Teaching Certificate Program <i>Northeastern University</i>	2024–2025
NRMN’s Research Mentor Training National Research Mentoring Network, <i>Northeastern University</i>	Summer 2024
ACL Year-Round Mentorship Association for Computational Linguistics, <i>Remote</i>	2022

SKILLS

Tools/Platforms: pyTorch, Huggingface, scikit-learn, statsmodels, Pandas, Azure, Google Cloud Platform (GCP), Docker, Weights & Biases, Prolific, Node.js, REST API, Flask, .NET Core

Front-end: React, Redux, Vue.js, Svelte, HTML5, CSS, SASS, Jekyll

Databases: MySQL, SQLite, PostgreSQL, MS SQL Server

Programming Languages: Python, Java, Javascript, C#, SQL, R, ~~W~~TeX, Shell, PHP, C

Human Languages: Korean and English (bilingual), Japanese (beginner), Swahili (novice)

Last updated: August 27, 2026